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Practical 4 Final Task B2

You are hired as a JavaScript developer for a Digital Locker System company. The company wants you to build a prototype program that secures user data (like PIN, secret notes), validates input, and provides some useful utility functions (like palindrome checks for secure PINs). The manager gave you the following requirements:

Part 1 – Functions, Scope, and Closures

1. Function Declarations

- Write a function getGreater(a, b) that takes two numbers and returns the greater number.
- This will be used to compare two entered PIN attempts and select the stronger one.

Input: getGreater(25, 48)

Output: 48

2. Function Expressions

- Write a function expression factorial that calculates the factorial of a given number.
- The factorial will be used to generate a unique hash code for secret storage.

Input: factorial(5)

Output: 120

3. Arrow Functions

- Create an arrow function `checkScript(str)` that checks if a given string contains the word "JavaScript".
- The Digital Locker will reject notes if they don't mention JavaScript at least once.

Input: `checkScript("I am learning JavaScript today!")`

Output: `true`

4. Closures

- Create a closure `secureMessage(password)` that stores a secret message.
- The inner function should return the message only if the given password matches; otherwise, return "Access Denied".

Input:

```
const locker = secureMessage("1234");  
console.log(locker("1111"));  
console.log(locker("1234"));
```

Output:

Access Denied

(Your secret message here)

5. Scope Check

Demonstrate global scope, function scope, and block scope in your program using meaningful variables like `lockerName`, `pin`, and `tempAccess`.

Part 2 – Error Handling & Palindrome Utilities

1. Reverse Number with Error Handling

- Write a function `reverseNumber(num)` that works for both positive and negative numbers.
- Use `try...catch` to throw an error if input is invalid.

Input: `reverseNumber(-1234)`

Output: `-4321`

2. Palindrome PIN Check

- Write a function `isPalindrome(num)` that checks whether a number is a palindrome.
- Use it to validate if a user's PIN is secure (palindromes are considered secure).

Input: `isPalindrome(1221)`

Output: `true`

3. Word Palindrome Check

- Extend functionality: accept a string input from the user and check if it is a palindrome (word-based).
- Ignore case and spaces (e.g., "Race car" → palindrome).

Input: `"Madam"`

Output: `true`

4. Square Root with Error Handling

- Write a function that takes a number and returns its square root.
- If input is invalid or negative, throw an error and handle it with try...catch.

Input: `sqrt(-9)`

Output: "Error: Negative numbers not allowed"

5. Prime Number Checker with Arrow Function

- Create an arrow function `isPrime(num)` that checks whether a number is prime.
- Use try...catch to handle invalid inputs (e.g., strings or negative numbers).

Input: `isPrime(7)`

Output: `true`

Final Integration Challenge

Put it all together in a menu-driven program simulation (console-based).

The program should ask the user to choose an option:

1. Compare two PINs and show the stronger one.
2. Generate a factorial hash for a number.
3. Check if a message contains "JavaScript".
4. Store and access a secret message (closure).
5. Reverse a number and check if it's a palindrome.
6. Check if a word is a palindrome.
7. Find square root with error handling.
8. Check if a number is prime.

For each option, display proper input prompts, error handling, and results.

Do NOT copy code directly from examples above.

You must adapt logic, rename variables, and integrate into the scenario.

If your code looks copy-pasted, it will not be accepted.

```
<> prac4final(1).html > html > body > script
1  <!DOCTYPE html>
2  <html>
3  <head>
4  | <title>Digital Locker System</title>
5  </head>
6  <body>
7  | <h2>Digital Locker Menu</h2>
8  | <div id="options">
9  |   <button onclick="comparePins()">1. Compare two PINs and show the stronger one</button><br><br>
10 |   <button onclick="generateFactorial()">2. Generate a factorial hash for a number</button><br><br>
11 |   <button onclick="checkMessage()">3. Check if a message contains "JavaScript"</button><br><br>
12 |   <button onclick="storeSecret()">4. Store and access a secret message (closure)</button><br><br>
13 |   <button onclick="reverseAndPalindrome()">5. Reverse a number and check if it is a palindrome</button><br><br>
14 |   <button onclick="checkWordPalindrome()">6. Check if a word is a palindrome</button><br><br>
15 |   <button onclick="findSquareRoot()">7. Find square root with error handling</button><br><br>
16 |   <button onclick="checkPrime()">8. Check if a number is prime</button><br><br>
17 | </div>
18 | <h3>Output:</h3>
19 | <pre id="output"></pre>
20 |
21 | <script>
22 |   let currentPin = 0;
23 |
24 |   function getHigherPin(x, y) {
25 |     if (typeof x !== 'number' || typeof y !== 'number') throw new Error('Invalid PIN');
26 |     let stronger = x >= y ? x : y;
27 |     currentPin = stronger;
28 |     return stronger;
29 |   }
30 |
31 |   const factorialHash = function(n) {
32 |     if (typeof n !== 'number' || n < 0 || !Number.isInteger(n)) throw new Error('Invalid input');
33 |     let res = 1;
34 |     for (let i = 2; i <= n; i++) res *= i;
35 |     return res;
36 |   };
37 |
38 |   const containsJavaScript = (str) => /javascript/i.test(str);
39 |
40 |   function createSecretVault(password) {
41 |     let secret = "This is your confidential note.";
42 |     return function(passAttempt) {
43 |       return passAttempt === password ? secret : "Access Denied";
44 |     };
45 |   }

```

```

47 function reverseNumber(num) {
48   if (typeof num !== 'number' || isNaN(num)) throw new Error('Invalid number');
49   let sign = Math.sign(num);
50   let rev = parseInt(Math.abs(num).toString().split('').reverse().join(''));
51   return sign * rev;
52 }
53
54 function isNumericPalindrome(num) {
55   let str = Math.abs(num).toString();
56   return str === str.split('').reverse().join('');
57 }
58
59 function isWordPalindrome(word) {
60   let clean = word.toLowerCase().replace(/\s+/g, '');
61   return clean === clean.split('').reverse().join('');
62 }
63
64 function safeSqrt(num) {
65   if (typeof num !== 'number' || isNaN(num)) throw new Error('Invalid number');
66   if (num < 0) throw new Error('Negative numbers not allowed');
67   return Math.sqrt(num);
68 }
69
70 const isPrime = (num) => {
71   if (typeof num !== 'number' || !Number.isInteger(num) || num < 2) throw new Error('Invalid input');
72   for (let i = 2; i <= Math.sqrt(num); i++) if (num % i === 0) return false;
73   return true;
74 };
75
76 const outEl = document.getElementById('output');
77 function log(msg) {
78   outEl.textContent += msg + "\n";
79 }
80
81 function comparePins() {
82   let a = parseInt(prompt("Enter first PIN:"));
83   let b = parseInt(prompt("Enter second PIN:"));
84   try {
85     let stronger = getHigherPin(a, b);
86     log("Stronger PIN: " + stronger);
87   } catch (e) {

```

```

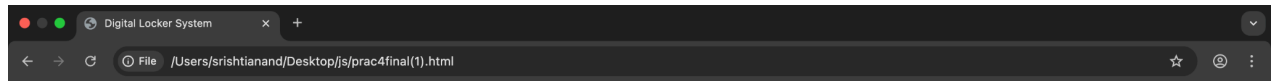
88     log(e.message);
89     }
90 }
91
92 function generateFactorial() {
93     let n = parseInt(prompt("Enter number for factorial hash:"));
94     try {
95         log("Factorial Hash: " + factorialHash(n));
96     } catch (e) {
97         log(e.message);
98     }
99 }
100
101 function checkMessage() {
102     let msg = prompt("Enter your note:");
103     log("Contains JavaScript: " + containsJavaScript(msg));
104 }
105
106 function storeSecret() {
107     let pwd = prompt("Set a password for your vault:");
108     let vault = createSecretVault(pwd);
109     let attempt = prompt("Enter password to access secret:");
110     log(vault(attempt));
111 }
112
113 function reverseAndPalindrome() {
114     let n = parseInt(prompt("Enter a number:"));
115     try {
116         let rev = reverseNumber(n);
117         log("Reversed Number: " + rev);
118         log("Is Palindrome: " + isNumericPalindrome(n));
119     } catch (e) {
120         log(e.message);
121     }
122 }
123
124 function checkWordPalindrome() {
125     let word = prompt("Enter a word or phrase:");
126     log("Is Palindrome: " + isWordPalindrome(word));
127 }
128

```

```

129 function findSquareRoot() {
130     let n = parseFloat(prompt("Enter number for square root:"));
131     try {
132         log("Square Root: " + safeSqrt(n));
133     } catch (e) {
134         log("Error: " + e.message);
135     }
136 }
137
138 function checkPrime() {
139     let n = parseInt(prompt("Enter number to check prime:"));
140     try {
141         log("Is Prime: " + isPrime(n));
142     } catch (e) {
143         log(e.message);
144     }
145 }
146 </script>
147 </body>
148 </html>
149

```



Digital Locker Menu

1. Compare two PINs and show the stronger one
2. Generate a factorial hash for a number
3. Check if a message contains "JavaScript"
4. Store and access a secret message (closure)
5. Reverse a number and check if it's a palindrome
6. Check if a word is a palindrome
7. Find square root with error handling
8. Check if a number is prime

Output:

```
Stronger PIN: 2280
Factorial Hash: 720
Contains JavaScript: true
This is your confidential note.
Reversed Number: 1221
Is Palindrome: true
Is Palindrome: true
Square Root: 8.94427190999916
Is Prime: false
```